Sean Grogan, PhD

514-207-0075 ◇ 802-448-0772
sean.grogan@gmail.com

SUMMARY

I am an operations researcher and applied mathematician specializing in optimization, logistics, planning, and quantitative policy analysis with a strong geospatial component. My research applies mathematical modeling, stochastic optimization, metaheuristics, and machine learning (clustering, classification, and dimensionality reduction) to socially consequential problems – including UAV routing for disaster reconnaissance and spatial analysis of election accessibility and voter participation. I have a strong record of interdisciplinary collaboration, working alongside academic, government, and industry partners to produce peer-reviewed publications and deployable analytical tools. I am committed to research that incorporates uncertainty explicitly into the decision-making process, producing solutions that are both robust and practically deployable. My technical toolkit includes Python (scikit-learn, PyTorch), R, C++, CPLEX, Gurobi, QGIS, and SQL, and I am eligible to work in Canada, the United States, and the European Economic Area.

EDUCATION

PhD, Industrial Engineering and Applied Mathematics 2023
Polytechnique Montréal
Thesis : A data-driven approach to locating and routing unmanned aerial vehicles for disaster area reconnaissance

MSc, Computer Science and Operations Research 2016
Université de Montréal
Thesis : An adaptive neighborhood search algorithm for optimizing stochastic mining complexes

BEng, Mining and Metallurgical Engineering Co-Op Program 2013
McGill University
Graduate Research Project : Stochastic optimization of a mine using simulated annealing

PROFESSIONAL EXPERIENCE

Manager, Business Intelligence 2024 — Present
Elections Ontario
Toronto, Ontario, Canada

Post Doctoral Fellow Feb 2023 — Apr 2024
Research Fellow
Laboratory of Elections and Democracy - Laboratoire sur les Élections et la Démocratie
Department of Political Science & Economics
Royal Military College of Canada, Kingston, Ontario, Canada

Research Engineer and Doctoral Candidate Sep 2016 — Apr 2023
Chaire de recherche Jarislowsky/SNC-Lavalin en gestion de projets internationaux (now AtkinsRéalis)
Département de Mathématiques et de Génie Industriel
Polytechnique Montréal, Montréal, Québec, Canada

Research Assistant Jan 2020 — Dec 2021
Laboratory of Elections and Democracy - Laboratoire sur les Élections et la Démocratie
Department of Political Science & Economics
Royal Military College of Canada, Kingston, Ontario, Canada

Research Assistant and M.Sc. Candidate 2014 — 2016
COSMO — Stochastic Mine Planning Laboratory
Department of Mining and Materials Engineering
McGill University, Montréal, Québec, Canada

Long Range Mine Planner – Intern Apr — Sep 2013
Newmont Mining Corp, Denver, Colorado, United States of America

Head Coach Sep 2011—2013
Dawson College Lacrosse Program, Montréal, Quebéc, Canada

Short Range Mine Planner – Intern Jan — Sep 2011
Syncrude Canada Ltd, Fort McMurray, Alberta, Canada

On-Field Official, Lacrosse and Instructor Seasonal 2009—2015
Crosse Quebéc, Quebéc, Canada

PUBLICATIONS

Journal Article : Garnett, Holly Ann; **Grogan, Sean** (2025). The last-minute voter: age and time of voting in Ontario, Canada. *Journal of Elections, Public Opinion and Parties* .

Journal Article : **Grogan, Sean**; Gamache, Michel; Pellerin, Robert (2023). Using historical data to dynamically route post-disaster assessment unmanned aerial vehicles in the context of responding to tornadoes. *Applied Sciences*.

Journal Article : **Grogan, Sean**; Pellerin, Robert; Gamache, Michel; Perrier, Nathalie (2022). Location of disaster assessment UAVs using historical tornado data. *Geomatics, Natural Hazards and Risk*.

Journal Article : **Grogan, Sean**; Pellerin, Robert; Gamache, Michel (2021). Using tornado-related weather data to route unmanned aerial vehicles to locate damage and victims. *OR Spectrum*.

Journal Article : Garnett, Holly Ann; **Grogan, Sean** (2021). I came, I saw, I voted: Distance to Polling Locations and Voter Turnout in Ontario, Canada. *Canadian Journal of Political Science*.

Conference Proceedings : **Grogan, Sean**; Gamache, Michel; & Pellerin, Robert (2019). Dynamically Routing UAVs in the Aftermath of a Severe Tornado. *In Edition ProLog 2019*. Metz, FR.

Conference Proceedings : **Grogan, Sean**; Gamache, Michel; & Pellerin, Robert (2018). The use of unmanned aerial vehicles and drones in search and rescue operations - a survey. *In Edition ProLog 2018*. Hull, UK.

Masters Thesis : **Grogan, Sean** (2016). An adaptive neighborhood search algorithm for optimizing stochastic mining complexes. *Université de Montréal*.

TEACHING EXPERIENCE

Manager, Business Intelligence 2024–Present
Toronto, Ontario, Canada

- Manager position, first manager of Business Intelligence at Elections Ontario.
- Established and built the Business Intelligence team from the ground up, through five hiring campaigns (up to three direct reports), including recruitment, hiring, on-boarding, training, and performance evaluations; team was productive from day one.
- Created on-boarding curriculum to educate new BI Analysts on the nature of the Permanent Register of Ontario, technical implementation, and legal context of the Register.

- Direct the team's execution of statutory reporting required under the Election Act, translating legal and regulatory requirements into data and reporting practice.
- Completed Hiring & Selection Training (Brock University, 2026) to formalize recruitment practices.

English Language and Technical Writing Tutor 2014–2023
Montréal, Quebéc, Canada

- English writing proofreader and tutor.
- Assisted peers in increasing clarity, conciseness, and expressing their technical knowledge in the English.
- Assured that the paper was in adherence to style guides, predominately APA and IEEE styles.

COSMO Mine Links Tutorial for COSMO Tech Day Annual Lecture Series 2015—2016
Montréal, Quebéc, Canada

- Multi-day conference and instruction session for using COSMO Mine Links mine optimization software for a diverse set of industry partners.
- Provided instruction on using software for the mineral optimization processes.
- Provided detailed information on optimization improvements of the software for the companies' mineral optimization processes.
- Conducted lectures and individual tutoring on using COSMO Mine Links

COSMO Mine Links Tutorial for Newmont Mining Corporation 2014
Winnemucca, Nevada, United States of America

- Multi-day instruction session for using COSMO Mine Links mine optimization software for Newmont Mining Corp.
- Incorporated mine site visits into improving the COSMO Mine Links mine optimization software.
- Conducted lectures and individual tutoring on using COSMO Mine Links

Head Coach, Dawson College Lacrosse Program Sep 2011—2013
Montréal, Quebéc, Canada

- Developed and implemented comprehensive lacrosse training programs for CEGEP-level athletes.
- Created practice plans and game plans to place student-athletes in the best positions for team success.
- Fostered a team culture promoting sportsmanship, deliberate practice, and respect on and off the field.
- Liaised with CEGEP administration, to communicate team progress, achievements, and upcoming events.

National Officials Lacrosse Certification Program, Instructor 2012—2014
Montréal, Quebéc, Canada

- Conducted training sessions for new and returning Lacrosse officials
- Sessions included classroom theory and on-field practice sessions.

AWARDS AND SCHOLARSHIPS

SSHRC Partnership Engage Grant February 2023
Laboratory of Elections and Democracy - Laboratoire sur les Élections et la Démocratie
Social Sciences and Humanities Research Council of Canada
Co-authored the successful grant proposal.

Jarislowsky Foundation Doctoral Fellowship September 2016
Polytechnique Montréal
Renewed Annually 2016 – 2023

Three Minute Thesis - Second Place March 2018
Polytechnique Montréal
3MT is a university wide competition in which participants present their research and its wider impact in 3 minutes or less to a panel of non-specialist judges.

NSERC Award — Sustainable Mineral Resource Development and Optimization Under Uncertainty January 2014
COSMO – Stochastic Mine Planning Laboratory, McGill University
Renewed By Semester 2014—2016

VOLUNTEER ACTIVITIES

Loyola High School Science Fair Volunteer Annually 2014 — 2023
Montréal, Quebéc, Canada

Judge, Montreal Regional Science Fair, CEGEP and Secondary School Annually 2020 — 2023
Montréal, Quebéc, Canada

Pro Bono Analytics, 10:12 Sports June 2020 — December 2020
Baltimore, Maryland, United States of America

PROLOG 2019 Project Logistic Scientific Committee and Track Coordinator June 2019
IAE Metz School of Management, Université de Lorraine, Metz, France

OTHER CONTRIBUTIONS

Computer Programming:

Computer Program : **Grogan, Sean** (2023). Great Circle Calculator. *Python 3*.

Data Science and Engineering:

Data Engineering : Garnett, Holly Ann (2024). Who Gives? Who Gets? Who Wins? Who Cares? Campaign Finance in Local Races. *University of Toronto Press*. (Provided technical expertise in processing Canadian census and Canadian federal campaign finance data.)

Data Engineering : Garnett, Holly Ann (2024). Where do donors come from? Using census data to predict donations to Canadian federal election candidates. *Political Geography*. (Provided technical expertise in processing Canadian census and Canadian federal campaign finance data.)

Data Engineering : MacGregor, Madison (2023). Public Perceptions of Elections Ontario in the 2022 Provincial Elections: A Twitter analysis. *2023 Forum on Democratic Citizenship*. (Provided technical expertise in pulling and initial processing of Twitter data.)

Other Contributions:

Internal Report: Garnett, Holly Ann; **Grogan, Sean** (2023). The Ontario Voter Experience: Understanding Ontario Voters. Internal report for Elections Ontario.

Internal Report: **Grogan, Sean** (2016). Metaheuristic Improvements to COSMO Mine Links. Internal report for Newmont Mining Corporation.

Editorial : **Grogan, Sean** (2013). Applying mathematical models to complex mining operations at Newmont *McGill Reporter*, Sept 19, 2013.

CONFERENCE PARTICIPATION

2023 Conference of Chief Electoral Officers July 2023
Victoria, British Columbia, Canada

2020 CPSA Annual Conference June 2020
University of Western Ontario, London, Ontario, Canada

2020 CSDC Conference – Democracy and Diversity April 2020
Centre for the Study of Democratic Citizenship, Montréal, Quebéc, Canada

PROLOG 2019 Project Management and Logistics June 2019
IAE Metz School of Management, Université de Lorraine, Metz, France

Optimization Days 2019 May 2019
HEC Montréal, Montréal, Quebéc, Canada

INFORMS Annual Meeting 2018 November 2018
Phoenix, Arizona, United States of America

PROLOG 2018 Project Managment and Logistics June 2018
University of Hull, Hull, England

Optimization Days 2018 May 2018
HEC Montréal, Montréal, Quebéc, Canada

COSMO Stochastic Mine Technical Days 2016 June 2016
McGill University, Montréal, Quebéc, Canada

COSMO Stochastic Mine Technical Days 2015 June 2015
McGill University, Montréal, Quebéc, Canada

CORS Annual Conference 2015 April 2015
Canadian Operational Research Society, Montréal, Quebéc, Canada

Integrated Strategic Planning Seminar 2014 May 2014
Toronto, Ontario, Canada

SKILLS AND RESEARCH INTERESTS

Skills	Computer Programming ; Mathematical Modeling ;
Research Domains	Operations Research ; Mathematical Modeling ; Stochastic Optimization ; Metaheuristics; Hyper-heuristics ; Mathematical Modeling with Uncertainty ; Machine Learning ; Spatial Data Analysis
Research Interests	Disaster Preparation ; Humanitarian Logistics ; Public Sector OR ; Geographic Information Systems ; Natural Resource Utilization ; Mine Planning ; Election Management
Software	Python (scikit-learn, PyTorch) ; R ; C++/C ; QGIS ; CPLEX ; GUROBI ; SQL
Languages	English ; French
Regions Eligible to Work	Canada, United States, European Economic Area

GROUPS, AFFILIATIONS, AND SOCIETIES

The Association of Municipal Managers, Clerks and Treasurers of Ontario (AMCTO)

Institute for Operations Research and the Management Sciences (INFORMS)
Canadian Operational Research Society (CORS)
Elections and Democracy Lab (LED)
Institute of Intergovernmental Relations
Chaire de recherche Jarislowsky/SNC-Lavalin en gestion de projets internationaux
Montreal Operations Research Student Chapter (MORSC)
Stochastic Mine Planning Laboratory at McGill University (COSMO)
Project Management Institute (PMI)
PMI - Montréal
Canadian Political Science Association (CPSA)
Centre interuniversitaire de recherche sur les réseaux d'entreprise, la logistique et le transport (CIRRELT)
Groupe d'études et de recherche en analyse de décisions (GERAD)
Groupe de recherche en gestion et mondialisation de la technologie (GMT)

ON THE INTERNET

GitHub (Personal Projects)	`https://github.com/seangrogan/`
GitHub (Research Material)	`https://github.com/seangrogan-researchprojects`
LinkedIn	`https://www.linkedin.com/in/seangrogan/`
ORCID	`https://orcid.org/0000-0002-1347-7874/`
Google Scholar	`https://scholar.google.ca/citations?user=7J9R0K0AAAAJ`
Web Site	`https://www.seangrogan.net/`

Erdős number : 5